

Power BI Data Analysis – Summary Notes

1. Getting Started

I downloaded the dataset from the attached link and imported it into Power BI.

2. Data Load into Power BI

I loaded the data into Power BI using the Get Data option. After loading the data, I cleaned and transformed it using Power Query Editor. Then, I created relationships between tables in the Model view. Finally, I created calculated columns and measures using DAX.

3. Data Model Overview

I used a star schema model where Sales_Fact is the main table. It is connected to dimension tables like Date_Dim, Region_Dim, and Customer_Dim. Date_Dim and Sales_Fact are connected using DateKey in a one-to-many relationship.

4. Return Flag

I created a calculated column called ReturnFlag. If the return value is blank, it is marked as 'Not Returned'. If a return value exists, it is marked as 'Returned'. This helps me analyze returned and non-returned sales easily.

5. Sales With and Without Filters

I created a Total Sales measure that shows sales with all filters applied. Using CALCULATE and ALL(Region_Dim), I removed region filters. This helped me compare regional sales with total company sales.

6. Region Contribution Percentage

I calculated Region Sales Percentage by dividing regional sales by total sales across all regions. This shows how much each region contributes to overall sales.

7. Year-over-Year and Month-over-Month Analysis

I used Quick Measures to calculate Year-over-Year growth and Month-over-Month sales difference. These calculations require a proper date field from Date_Dim and at least two years or months of data.

8. DateKey and Date_Dim Usage

The Sales_Fact table does not store actual dates, it only contains DateKey. All date-related calculations use Date_Dim[Date]. I used RELATED() only in calculated columns when date information was required inside the Sales_Fact table.

9. Common DAX Functions

I used basic DAX functions such as SUM, AVERAGE, and MAX. DISTINCTCOUNT and COUNTX helped me count unique and conditional values. IF, AND, OR, and SWITCH were used for conditional logic.

10. Text and Date Functions

I used text functions like CONCATENATE, UPPER, and LEFT to clean and format text. For date handling, I used YEAR, MONTH, and EOMONTH through the Date_Dim table.

11. Time Intelligence Functions

I used TOTALYTD to calculate year-to-date sales. SAMEPERIODLASTYEAR helped me compare sales with the previous year. DATESINPERIOD was used for rolling time analysis such as the last three months.

12. Running Total

I created a running total using CALCULATE and DATESBETWEEN. This shows cumulative sales over time and does not reset automatically each year.

13. Iterator Functions (SUMX and AVERAGEX)

I used SUMX for row-by-row calculations such as Quantity multiplied by Unit Price. AVERAGEX was used to calculate averages like average monthly sales and average sales per customer.

14. Key Learnings

- I learned to use Date_Dim for all time-based analysis
- I used ALL() to remove filters for comparison
- Time intelligence functions need proper date context
- Iterator functions work row by row
- Matrix visuals helped me validate my calculations